

Maintenance Engineering and Practices

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

5113D

Credits

4

Periods

5/week

Course Details

Title: Maintenance Engineering and Practices

Department: Common

Revision: REV_Maintenance Engineering and Practices

Pattern: Theory / Practical

Complies with official SITTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 5113D — Maintenance Engineering and Practices**. Use the official SITTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

[illegible]

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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Full Answer Key

OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Maintenance Engineering and Practices.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3, CO2 → PO2: 3, CO3 → PO3: 3, CO4 → PO3: 3.

MODULE 1

Core Principles

Outcomes

Master foundational terminology and core definitions of Maintenance Engineering and Practices.

Study

Detailed analysis of core principles and theoretical foundations.

Malayalam Support Note: ുറുറുറുറു ുറുറുറുറുറുറുറുറുറുറു ുറുറുറുറുറു ുറുറുറുറുറുറുറുറു
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MODULE 2

Applied Methods

Outcomes

Apply standard calculation techniques and operational procedures.

Study

Numerical problem-solving techniques and practical workflows.

Malayalam Support Note: ുറുറുറുറുറുറു ുറുറുറുറുറുറുറുറുറുറുറുറുറുറുറുറു ുറുറുറുറുറു
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MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: ഏകീകരണത്തിന്റെയും പ്രവർത്തനത്തിന്റെയും സങ്കീർണ്ണമായ വിലയിരുത്തലും മെച്ചപ്പെടുത്തലും. സങ്കീർണ്ണമായ സിസ്റ്റങ്ങളെ വിലയിരുത്തുകയും പ്രവർത്തനം മെച്ചപ്പെടുത്തുകയും ചെയ്യുക.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: സമഗ്ര സാങ്കേതിക പരിഹാരങ്ങൾ രൂപകല്പന ചെയ്യുകയും കൈവശപ്പെടുത്തൽ പ്രദർശിപ്പിക്കുകയും ചെയ്യുക. യഥാർത്ഥ കേസുകളും വ്യവസ്ഥാപിത പ്രോജക്ടുകളുടെ രൂപകല്പനയും.

Practice Model Question Paper

Time: 3 Hours | Max Marks: 75

Part A (2 marks each)

1. Define core concepts of Maintenance Engineering and Practices.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.